

Introduction to Binary Coded Decimal

Baseline Constraints

System utilizes 4 bits per digit.

Represents exact decimal values directly in binary format.

Valid Range:	0 to 9.
Valid Binary:	0000 to 1001.
Invalid Range:	10 to 15.
Invalid Binary:	1010 to 1111.

Initial Binary Addition

Step 1: Process adds 2 operands.

Step 2: Addition executes using 1 4-bit binary adder.

Step 3: Architecture enforces an absolute upper bound limit.

Maximum Possible Sum is 19

Boundary Condition: 9 (Operand A) + 9 (Operand B) + 1 (Input Carry) = 19

Error Detection Logic

$$C = K + Z_8Z_4 + Z_8Z_2$$

Trigger Condition A:
Sum > 9.

Trigger Condition B:
Carry = 1 (K = 1).

K represents the carry out from the 4-bit adder.
Z₈, Z₄, Z₂ represent the sum bits from the 4-bit adder.

Correction Stage

Problem Definition

Condition: Result resides in the invalid range (**10** to **19**).

Mechanical Fix

Action: Add **6** to the invalid result.

Binary Constant: **0110**.

Hardware Requirement

Implementation requires **1** additional **4**-bit adder to process the correction stage.

Resulting Output generates the valid decimal digit and passes **1** carry to the next stage.

Synthesis Application Example

Inputs: Adding 5 and 7.

Initial Result: Sum is 12 (Binary **1100**).

Logic Check: Sum > 9 triggers the error detection logic.

Correction: Adding **6** (Binary **0110**) to the intermediate result.

Final Output: Gives correct BCD format.

Final Result: **0001 0010**.